Python and PyPSA for Energy Modelling: Learning Resources

This document lists the resources that support the workshop notebook and further study in energy system modelling. The first table maps each section of the notebook to a relevant help page; the remaining sections are for further study. All links are to official documentation, university courses or peer-reviewed publications, and were checked on 12 October 2026.

Support for the workshop notebook

When an error occurs, read the last line of the error message first, as it identifies the problem. The message `NameError: name '____' is not defined` indicates that a blank has not yet been completed. After correcting a cell that defines a function, run that cell again before re-running the test.

Notebook section	If you are stuck on	Read
1, 2 Setup and Python basics	Variables, loops, `if`, functions	The Python Tutorial, sections 3 to 4 (Python Software Foundation)
2, 3 Python basics and data	DataFrames, columns, selecting rows	10 minutes to pandas
3.2 Solar data	What PVGIS returns and its settings	PVGIS API (non-interactive service) (EC Joint Research Centre)
3.3 Demand profile	UTC, GMT and BST in pandas	Time zone handling
3.5, 4.7 Monthly totals	Hourly data to monthly sums	`Series.resample`
3.5 to 5.7 Figures	Axes, labels, saving plots	Matplotlib quick start
4.3 Battery rule	Arrays and element-wise maths	NumPy: the absolute basics
4.1 Measures	Self-consumption and self-sufficiency rates	Luthander et al. (2015), in Key papers below
5.3 PyPSA model	Buses, loads, generators	PyPSA components
5.3 to 5.4 Solving	`n.optimize()` and its options	`n.optimize` reference and optimisation overview
5.5 Prices	Merit order and clearing price in PyPSA	Quickstart 1: Markets and simple electricity markets
Any	Running and sharing notebooks	Opening JupyterLab from Anaconda Navigator (Anaconda), The JupyterLab interface (Project Jupyter) and Welcome to

Notebook section	If you are stuck on	Read
		Colab (Google)

Learn Python for energy data

All resources are freely available online. Those new to Python are advised to work through them in the order listed.

Resource	Provider	What it covers
Programming with Python	Software Carpentry	First steps: variables, lists, loops, functions, with scientific data
Plotting and Programming in Python	Software Carpentry	pandas and plotting for beginners
The Python Tutorial	Python Software Foundation	The official reference tutorial for the language
Python for Data Analysis, 3rd edn	Wes McKinney, creator of pandas	pandas in depth, including time series
Python Data Science Handbook	Jake VanderPlas	NumPy, pandas, Matplotlib and an introduction to machine learning

PyPSA and open modelling tools

PyPSA is the modelling framework used in Task 2. Start with the three Quickstart notebooks, then pick the examples closest to your project.

Resource	What it is for
PyPSA documentation	Official user guide, examples and API reference
Installation	Installing PyPSA and a solver on your own computer
Quickstart 1: Markets	Dispatch and prices, the closest match to Task 2
Quickstart 2: Power flow	Networks with lines and power flow
Quickstart 3: Investments and storage	Choosing what to build, including batteries
Single-node capacity expansion	Least-cost mix of wind, solar and storage for one region
Unit commitment	Start-up costs and minimum output, missing from Task 2
All examples	The full gallery of example notebooks

Resource	What it is for
PyPSA on GitHub	Source code, issues and Discussions for questions
PyPSA-Eur	Open model of the European energy system, including Great Britain
Linopy	The optimisation layer PyPSA builds on
HiGHS	The open-source solver used in the notebook
pvlib python	Detailed solar PV modelling in Python
Open Energy Modelling Initiative	Community of open energy modellers, with a forum

Courses and lecture notes

- Data Science for Energy System Modelling, Dr Fabian Neumann, TU Berlin. A free course with notebooks that takes you from pandas to PyPSA. Recommended as the next step after this workshop.
- Energy Systems lecture notes, Prof Tom Brown, TU Berlin. Lecture slides on electricity markets, optimisation, storage and networks; the teaching page lists earlier years.
- Kirschen, D.S. and Strbac, G. (2018) *Fundamentals of Power System Economics*. 2nd edn. Chichester: Wiley. ISBN 978-1-119-21324-6. The standard textbook on merit order, market clearing and pricing; available through most university libraries.

Open data for UK energy modelling

Source	Provider	Data
PVGIS (interactive tool)	European Commission Joint Research Centre	Modelled hourly solar PV output and irradiance; used in Task 1
Renewables.ninja	Pfenninger and Staffell	Simulated hourly wind and solar output for any location
NESO Data Portal	National Energy System Operator	GB demand, generation mix, carbon intensity and forecasts
Elexon Insights (API)	Elexon	GB balancing and settlement data, generation by fuel type
Open Power System Data	Open Power System Data project	European time series and power plant lists; no longer updated since 2020

Key papers

Brown et al. (2018a) is the recommended starting point for PyPSA, and Luthander et al. (2015) for Task 1. Where a publisher requests a login, access is usually available through your university library.

1. Brown, T., Hörsch, J. and Schlachtberger, D. (2018a) 'PyPSA: Python for Power System Analysis', *Journal of Open Research Software*, 6(1), p. 4. doi:10.5334/jors.188
2. Brown, T., Schlachtberger, D., Kies, A., Schramm, S. and Greiner, M. (2018b) 'Synergies of sector coupling and transmission reinforcement in a cost-optimised, highly renewable European energy system', *Energy*, 160, pp. 720-739. doi:10.1016/j.energy.2018.06.222
3. Hofmann, F. (2023) 'Linopy: Linear optimization with n-dimensional labeled variables', *Journal of Open Source Software*, 8(84), p. 4823. doi:10.21105/joss.04823
4. Holmgren, W.F., Hansen, C.W. and Mikofski, M.A. (2018) 'pvlib python: a python package for modeling solar energy systems', *Journal of Open Source Software*, 3(29), p. 884. doi:10.21105/joss.00884
5. Hörsch, J., Hofmann, F., Schlachtberger, D. and Brown, T. (2018) 'PyPSA-Eur: An open optimisation model of the European transmission system', *Energy Strategy Reviews*, 22, pp. 207-215. doi:10.1016/j.esr.2018.08.012
6. Huangfu, Q. and Hall, J.A.J. (2018) 'Parallelizing the dual revised simplex method', *Mathematical Programming Computation*, 10(1), pp. 119-142. doi:10.1007/s12532-017-0130-5
7. Luthander, R., Widén, J., Nilsson, D. and Palm, J. (2015) 'Photovoltaic self-consumption in buildings: A review', *Applied Energy*, 142, pp. 80-94. doi:10.1016/j.apenergy.2014.12.028
8. Pfenninger, S., DeCarolis, J., Hirth, L., Quoilin, S. and Staffell, I. (2017) 'The importance of open data and software: Is energy research lagging behind?', *Energy Policy*, 101, pp. 211-215. doi:10.1016/j.enpol.2016.11.046
9. Pfenninger, S., Hawkes, A. and Keirstead, J. (2014) 'Energy systems modeling for twenty-first century energy challenges', *Renewable and Sustainable Energy Reviews*, 33, pp. 74-86. doi:10.1016/j.rser.2014.02.003
10. Pfenninger, S. and Staffell, I. (2016) 'Long-term patterns of European PV output using 30 years of validated hourly reanalysis and satellite data', *Energy*, 114, pp. 1251-1265. doi:10.1016/j.energy.2016.08.060